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associate professor

基本信息

姓名: KOBAYASHI Masaharu

所属院校: 东京大学

地区: Japan

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

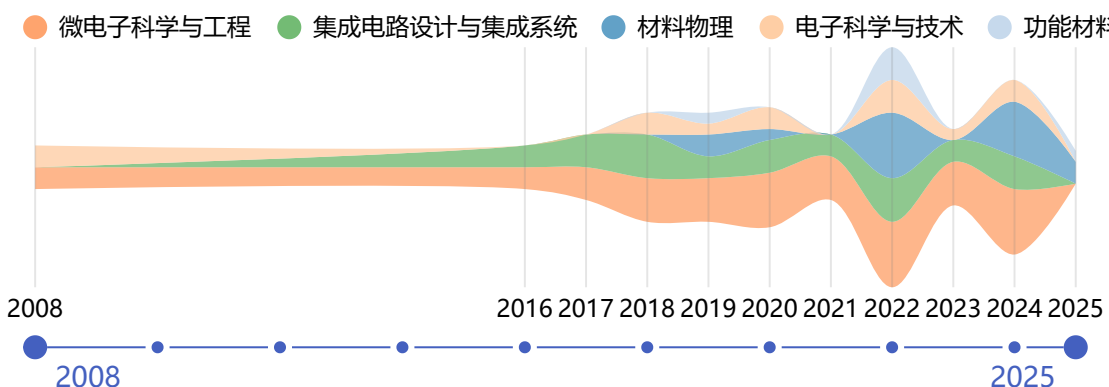
存储器设计与架构

低功耗电路与能量管理

个人简介

KOBAYASHI Masaharu is an Associate Professor in the Department of Informatics and Electronics at the Graduate School of Engineering, specializing in Integrated Nanoelectronics. His academic focus is on Electrical Engineering and Information Systems.

近年发表的论文



Structural and electronic properties of interstitial oxygen defect in amorphous indium gallium oxide (IGO) semiconductors: a theoretical study

JAPANESE JOURNAL OF APPLIED PHYSICS

Pandy, Chitra; Kobayashi, Masaharu

被引用次数: 0

2025

Enhanced reliability of ferroelectric Hf_{0.5}Zr_{0.5}O₂ capacitors by bottom electrode surface oxidation

JAPANESE JOURNAL OF APPLIED PHYSICS

Itoya, Yuki; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 0

2025

Performance and reliability of nanosheet oxide semiconductor FETs with ALD-grown InGaO for 3D integration (Invited)

2024 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM, IRPS 2024

Kobayashi, Masaharu; Hikake, Kaito; Li, Zhuo; Hao, Junxiang; Pandy, Chitra; Saraya, Takuya; Hiramoto, Toshiro; Takahashi, Takanori; Uenuma, Mutsunori; Uraoka, Yukiharu

被引用次数: 0

2024

A Nanosheet Oxide Semiconductor FET Using ALD InGaO_x Channel for 3-D Integrated Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hikake, Kaito; Li, Zhuo; Hao, Junxiang; Pandy, Chitra; Saraya, Takuya; Hiramoto, Toshiro; Takahashi, Takanori; Uenuma, Mutsunori; Uraoka, Yukiharu; Kobayashi, Masaharu

被引用次数: 3

2024

Optimized operation scheme of flash-memory-based neural network online training with ultra-high endurance

JOURNAL OF SEMICONDUCTORS

Feng, Yang; Sun, Zhaohui; Qi, Yueran; Zhan, Xuepeng; Zhang, Junyu; Liu, Jing; Kobayashi, Masaharu; Wu, Jixuan; Chen, Jiezhi

被引用次数: 0

2024

Innovative Recovery Strategy for MFIS-FeFETs at Optimal Timing With Robust Endurance: Fast-Unipolar Pulsing (100 ns), Nearly Zero Memory Window Loss (0.02%), and Self-Tracking Circuit Design

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wu, Cheng-Hung; Liu, Jay; Zheng, Xun-Ting; Chuang, Han-Fu; Tseng, Yi-Ming; Kobayashi, Masaharu; Su, Chun-Jung; Hu, Vita Pi-Ho

被引用次数: 1

2024

High-Field Transport and Statistical Variability of Nanosheet Oxide Semiconductor FETs With Channel Length Scaling

IEEE TRANSACTIONS ON ELECTRON DEVICES

Huang, Xingyu; Hikake, Kaito; Kim, Sung-Hun; Sakai, Kota; Li, Zhuo; Mizutani, Tomoko; Saraya, Takuya; Hiramoto, Toshiro; Takahashi, Takanori; Uenuma, Mutsunori; Uraoka, Yukiharu; Kobayashi, Masaharu

Transistor Matrix Array for Measuring Variability and Random Telegraph Noise at Cryogenic Temperatures

2024 IEEE 36TH INTERNATIONAL CONFERENCE ON MICROELECTRONIC TEST STRUCTURES, ICMTS 2024

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Oka, Hiroshi; Mori, Takahiro; Kobayashi, Masaharu; Hiramoro, Toshiro

被引用次数: 1

2024

Dielectric breakdown behavior of ferroelectric HfO₂ capacitors by constant voltage stress studied by in situ laser-based photoemission electron microscopy

JAPANESE JOURNAL OF APPLIED PHYSICS

Itoya, Yuki; Fujiwara, Hirokazu; Bareille, Cedric; Shin, Shik; Taniuchi, Toshiyuki; Kobayashi, Masaharu

被引用次数: 1

2024

Oxide-semiconductor channel ferroelectric field-effect transistors for high-density memory applications: 3D NAND operation and the potential impact of in-plane polarization

JAPANESE JOURNAL OF APPLIED PHYSICS

Hao, Junxiang; Mei, Xiaoran; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 0

2024

Cryogenic threshold voltage and on-current variability comparative analysis of same-fab 65 nm bulk and fully depleted silicon-on-insulator metal-oxide-semiconductor field-effect transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Liu, Zihao; Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Oka, Hiroshi; Mori, Takahiro; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2024

A Nanosheet Oxide Semiconductor FET Using ALD InZnOx Channel

2024 IEEE SILICON NANOELECTRONICS WORKSHOP, SNW 2024

Kim, Sung-hun; Hikake, Kaito; Li, Zhuo; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

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2024

Energy-Efficient Annealing Process of Ferroelectric Hf_{0.5}Zr_{0.5}O₂ Capacitor Using Ultraviolet-LED for Green Manufacturing

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Yamada, Hirotaka; Furue, Satoru; Yokomori, Takehiko; Itoya, Yuki; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 0

2024

Pixel-Parallel Three-Layer Stacked CMOS Image Sensors Using Double-Sided Hybrid Bonding of SOI Wafers

IEEE TRANSACTIONS ON ELECTRON DEVICES

Goto, Masahide; Honda, Yuki; Nanba, Masakazu; Iguchi, Yoshinori; Saraya, Takuya; Kobayashi, Masaharu; Higurashi, Eiji; Toshiyoshi, Hiroshi; Hiramoto, Toshiro

Effectiveness of c-Axis Aligned Crystalline IGZO FET as Selector Element and Ferroelectric Capacitor Scaling of 1T1C FeRAM

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Yazaki, Naomi; Motoyoshi, Ryosuke; Numata, Shiyu; Ohshima, Kazuaki; Egi, Yuji; Isaka, Fumito; Ohno, Toshikazu; Tezuka, Sachiaki; Hamada, Toshiki; Furutani, Kazuma; Tsuda, Kazuki; Matsuzaki, Takanori; Onuki, Tatsuya; Murakawa, Tsutomu; Kunitake, Hitoshi; Kobayashi, Masaharu; Yamazaki, Shunpei

被引用次数: 3

2023

MOSFET series resistance extraction at cryogenic temperatures

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Oka, Hiroshi; Mori, Takahiro; Kobayashi, Masaharu; Hiramoto, Toshiro

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2023

Device modeling of oxide-semiconductor channel antiferroelectric FETs using half-loop hysteresis for memory operation

JAPANESE JOURNAL OF APPLIED PHYSICS

Huang, Xingyu; Itoya, Yuki; Li, Zhuo; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

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2023

Ultrathin MoS₂-Channel FeFET Memory With Enhanced Ferroelectricity in HfZrO₂ and Body-Potential Control

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Xiang, Jiawen; Chang, Wen Hsin; Saraya, Takuya; Hiramoto, Toshiro; Irisawa, Toshifumi; Kobayashi, Masaharu

被引用次数: 7

2022

Efficient Erase Operation by GIDL Current for 3D Structure FeFETs With Gate Stack Engineering and Compact Long-Term Retention Model

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Mo, Fei; Xiang, Jiawen; Mei, Xiaoran; Sawabe, Yoshiki; Saraya, Takuya; Hiramoto, Toshiro; Su, Chun-Jung; Hu, Vita Pi-Ho; Kobayashi, Masaharu

被引用次数: 5

2022

A Threshold Voltage Definition Based on a Standardized Charge Versus Voltage Relationship

IEEE TRANSACTIONS ON ELECTRON DEVICES

Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 3

2022

Mesoscopic-scale grain formation in HfO₂-based ferroelectric thin films and its impact on electrical characteristics

NANO CONVERGENCE

Kobayashi, Masaharu; Wu, Jixuan; Sawabe, Yoshiki; Takuya, Saraya; Hiramoto, Toshiro

被引用次数: 25

2022

On the thickness dependence of the polarization switching kinetics in HfO₂-based ferroelectric

APPLIED PHYSICS LETTERS

Sawabe, Yoshiki; Saraya, Takuya; Hiramoto, Toshiro; Su, Chun-Jung; Hu, Vita Pi-Ho; Kobayashi, Masaharu

被引用次数: 19

2022

A robust single device MOSFET series resistance extraction method considering horizontal-field-dependent mobility (vol 61, SC1016, 2022)

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2022

Cause analysis of width-dependence of on-current variability in thin gate-all-around silicon nanowire MOSFET

JAPANESE JOURNAL OF APPLIED PHYSICS

Liu, Zihao; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2022

Monolithic 3D Integration of Oxide Semiconductor FETs and Memory Devices for AI Acceleration

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Kobayashi, Masaharu

被引用次数: 0

2022

Monolithic 3D Integration of Oxide Semiconductor FETs and Memory Devices for AI Acceleration (Invited)

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Kobayashi, Masaharu

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2022

Temperature-dependent Defect Behaviors in Ferroelectric Hf_{0.5}Zr_{0.5}O₂ Thin Film: Re-wakeup Phenomenon and Underlying Mechanisms

2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM

Li, Xiaopeng; Wu, Jixuan; Tai, Lu; Wei, Wei; Sang, Pengpeng; Feng, Yang; Chen, Bo; Zhao, Guoqing; Zhan, Xuepeng; Wang, Xiaolei; Kobayashi, Masaharu; Chen, Jiezhi

被引用次数: 8

2022

Effect of percolation path on temperature dependence of threshold voltage variability in bulk MOSFETs

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Oka, Hiroshi; Mori, Takahiro; Kobayashi, Masaharu; Hiramoto, Toshiro

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2022

A robust single device MOSFET series resistance extraction method considering horizontal-field-dependent mobility

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2022

A simulation study on memory characteristics of InGaZnO-channel ferroelectric FETs with 2D planar and 3D structures

JAPANESE JOURNAL OF APPLIED PHYSICS

Mo, Fei; Mei, Xiaoran; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 15

2022

Estimation of minimum operating voltage in fully depleted SOI SRAM cells using gamma distribution

JAPANESE JOURNAL OF APPLIED PHYSICS

Yu, Hongkuan; Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2022

Experimental investigations on ferroelectric dielectric breakdown in sub-10 nm Hf_{0.5}Zr_{0.5}O₂ film through comprehensive TDDb characterizations

JAPANESE JOURNAL OF APPLIED PHYSICS

Li, Xiaopeng; Wei, Wei; Wu, Jixuan; Tai, Lu; Zhan, Xuepeng; Zhang, Weiqiang; Tang, Mingfeng; Zhao, Guoqing; Xu, Hao; Chai, Junshuai; Wang, Xiaolei; Kobayashi, Masaharu; Chen, Jiezh

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2022

Variability characteristics and corner effects of gate-all-around (GAA) p-type poly-Si junctionless nanowire/nanosheet transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Ahn, Min-Ju; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2021

Analysis of Drain Current Variability Components in Extremely Narrow GAA Silicon Nanowire MOSFETs of 4nm Width

2021 SILICON NANO-ELECTRONICS WORKSHOP (SNW)

Liu, Zihao; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

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2021

Subthreshold Swing in Silicon Gate-All-Around Nanowire and Fully Depleted SOI MOSFETs at Cryogenic Temperature

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Sekiguchi, Shohei; Ahn, Min-Ju; Mizutani, Tomoko; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 7

2021

Design space exploration of hysteretic negative capacitance ferroelectric FETs based on static solutions of Landau-Khalatnikov model for nonvolatile memory applications

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 0

2021

Low-Voltage Operating Ferroelectric FET with Ultrathin IGZO Channel for High-Density Memory Application

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Mo, Fei; Tagawa, Yusaku; Jin, Chengji; Ahn, MinJu; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 86

2020

Physical Mechanisms of Reverse DIBL and NDR in FeFETs With Steep Subthreshold Swing

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Jin, Chengji; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 19

2020

A first-principles study on ferroelectric phase formation of Si-doped HfO₂ through nucleation and phase transition in thermal process

APPLIED PHYSICS LETTERS

Wu, Jixuan; Mo, Fei; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 34

2020

Superior subthreshold characteristics of gate-all-around p-type junctionless poly-Si nanowire transistor with ideal subthreshold slope

JAPANESE JOURNAL OF APPLIED PHYSICS

Ahn, Min-Ju; Saraya, Takuya; Kobayashi, Masaharu; Sawamoto, Naomi; Ogura, Atsushi; Hiramoto, Toshiro

被引用次数: 9

2020

Statistical analysis of temperature dependence of worst case static random access memory data retention voltage using extreme value theory

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 1

2020

A simulation study on low voltage operability of hafnium oxide based ferroelectric FET memories

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 5

2020

Width dependence of drain current and carrier mobility in gate-all-around multi-channel polycrystalline silicon nanowire transistors with 10nm width scale

JAPANESE JOURNAL OF APPLIED PHYSICS

Jang, Ki-Hyun; Saraya, Takuya; Kobayashi, Masaharu; Sawamoto, Naomi; Ogura, Atsushi; Hiramoto, Toshiro

被引用次数: 3

2020

Reduced variability of drain-induced barrier lowering and subthreshold slope at high temperature in bulk and silicon-on-thin-buried-oxide (SOTB) MOSFETs

JAPANESE JOURNAL OF APPLIED PHYSICS

Gao, Shuang; Mizutani, Tomoko; Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 4

2019

Solid State Devices and Materials

JAPANESE JOURNAL OF APPLIED PHYSICS

Hosoi, Takuji; Yaguchi, Hiroyuki; Kageshima, Hiroyuki; Chikamatsu, Masayuki; Fujiwara, Hirokazu; Furuta, Mamoru; Hirama, Kazuyuki; Johguchi, Kou; Kato, Toshiaki; Kawaguchi, Kenichi; Kikuchi, Akihiko; Kinoshita, Kentaro; Kitada, Hideki; Kobayashi, Masaharu; Kuroda, Rihito; Kuroki, Shinichiro; Matsudai, Tomoko; Minari, Takeo; Morioka, Hiroshi; Nagashio, Kosuke; Nakada, Kazuyoshi; Nakatsuka, Osamu; Oiwa, Akira; Okada, Hiroyuki; Okuda, Takafumi; Ono, Teruo; Sakamoto, Toshitsugu; Shiojima, Kenji; Shirao, Mizuki; Takeyama, Mayumi; Tanaka, Tetsu; Tatsuoka, Hirokazu; Tawara, Takehiko; Tayagaki, Takeshi; Tokuda, Takashi; Tsuda, Kunio; Ueki, Makoto; Yamamoto, Kazuhiko; Yeh, Wenchang

被引用次数: 0

2019

A Feasibility Study on Ferroelectric Shadow SRAMs Based on Variability-Aware Design Optimization

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

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2019

Ferroelectric HfO₂ Tunnel Junction Memory With High TER and Multi-Level Operation Featuring Metal Replacement Process

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Kobayashi, Masaharu; Tagawa, Yusaku; Mo, Fei; Saraya, Takuya; Hiramoto, Toshiro

被引用次数: 94

2019

On the Physical Mechanism of Transient Negative Capacitance Effect in Deep Subthreshold Region

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Jin, Chengji; Saraya, Takuya; Hiramoto, Toshiro; Kobayashi, Masaharu

被引用次数: 28

2019

Triple-Stacked Au/SiO₂ Hybrid Bonding With 6- μ m-Pitch Au Electrodes on Silicon-on-Insulator Substrates Using O₂ Plasma Surface Activation for 3-D Integration

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Honda, Yuki; Goto, Masahide; Watabe, Toshihisa; Hagiwara, Kei; Nanba, Masakazu; Iguchi, Yoshinori; Saraya, Takuya; Kobayashi, Masaharu; Higurashi, Eiji; Toshiyoshi, Hiroshi; Hiramoto, Toshiro

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2019

A perspective on steep-subthreshold-slope negative-capacitance field-effect transistor

APPLIED PHYSICS EXPRESS

Kobayashi, Masaharu

被引用次数: 66

2018

Experimental Observation and Simulation Model for Transient Characteristics of Negative-Capacitance in Ferroelectric HfZrO₂ Capacitor

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Jang, Kyungmin; Ueyama, Nozomu; Kobayashi, Masaharu; Hiramoto, Toshiro

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2018

Experimental Demonstration of a Nonvolatile SRAM With Ferroelectric HfO₂ Capacitor for Normally Off Application

IEEE JOURNAL OF THE ELECTRON DEVICES SOCIETY

Kobayashi, Masaharu; Ueyama, Nozomu; Jang, Kyungmin; Hiramoto, Toshiro

被引用次数: 11

2018

Role of gate current and polarization switching in sub-60 mV/decade steep subthreshold slope in metal-ferroelectric HfZrO₂-metal-insulator-Si FET

JAPANESE JOURNAL OF APPLIED PHYSICS

Jang, Kyungmin; Kobayashi, Masaharu; Hiramoto, Toshiro

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2018

Lowering data retention voltage in static random access memory array by post fabrication self-improvement of cell stability by multiple stress application

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 2

2018

Optimizing MOS-gated thyristor using voltage-based equivalent circuit model for designing steep-subthreshold-slope PN-body-tied silicon-on-insulator FET

JAPANESE JOURNAL OF APPLIED PHYSICS

Ueda, Daiki; Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 4

2018

On gate stack scalability of double-gate negative-capacitance FET with ferroelectric HfO₂ for energy efficient sub-0.2V operation

JAPANESE JOURNAL OF APPLIED PHYSICS

Jang, Kyungmin; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 2

2018

Measurement of Static Random Access Memory Power-Up State Using an Addressable Cell Array Test Structure

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Takeuchi, Kiyoshi; Mizutani, Tomoko; Shinohara, Hirofumi; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 2

2017

Negative Capacitance for Boosting Tunnel FET performance

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Kobayashi, Masaharu; Jang, Kyungmin; Ueyama, Nozomu; Hiramoto, Toshiro

被引用次数: 88

2017

Correlation between static random access memory power-up state and transistor variation

JAPANESE JOURNAL OF APPLIED PHYSICS

Takeuchi, Kiyoshi; Mizutani, Tomoko; Saraya, Takuya; Shinohara, Hirofumi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 6

2017

Parallel programmable nonvolatile memory using ordinary static random access memory cells

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Takeuchi, Kiyoshi; Saraya, Takuya; Shinohara, Hirofumi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 2

2017

Ion/Ioff ratio enhancement and scalability of gate-all-around nanowire negative-capacitance FET with ferroelectric HfO₂

SOLID-STATE ELECTRONICS

Jang, Kyungmin; Saraya, Takuya; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 29

2017

Transistor-level characterization of static random access memory bit failures induced by random telegraph noise

JAPANESE JOURNAL OF APPLIED PHYSICS

Mizutani, Tomoko; Saraya, Takuya; Takeuchi, Kiyoshi; Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 3

2016

On device design for steep-slope negative-capacitance field-effect-transistor operating at sub-0.2V supply voltage with ferroelectric HfO₂ thin film

AIP ADVANCES

Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 88

2016

On the origin of negative differential conductance in ultranarrow-wire-channel silicon single-electron and single-hole transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Kobayashi, Masaharu; Miyaji, Kousuke; Hiramoto, Toshiro

被引用次数: 15

2008

Experimental study on quantum confinement effects in silicon nanowire metal-oxide-semiconductor field-effect transistors and single-electron transistors

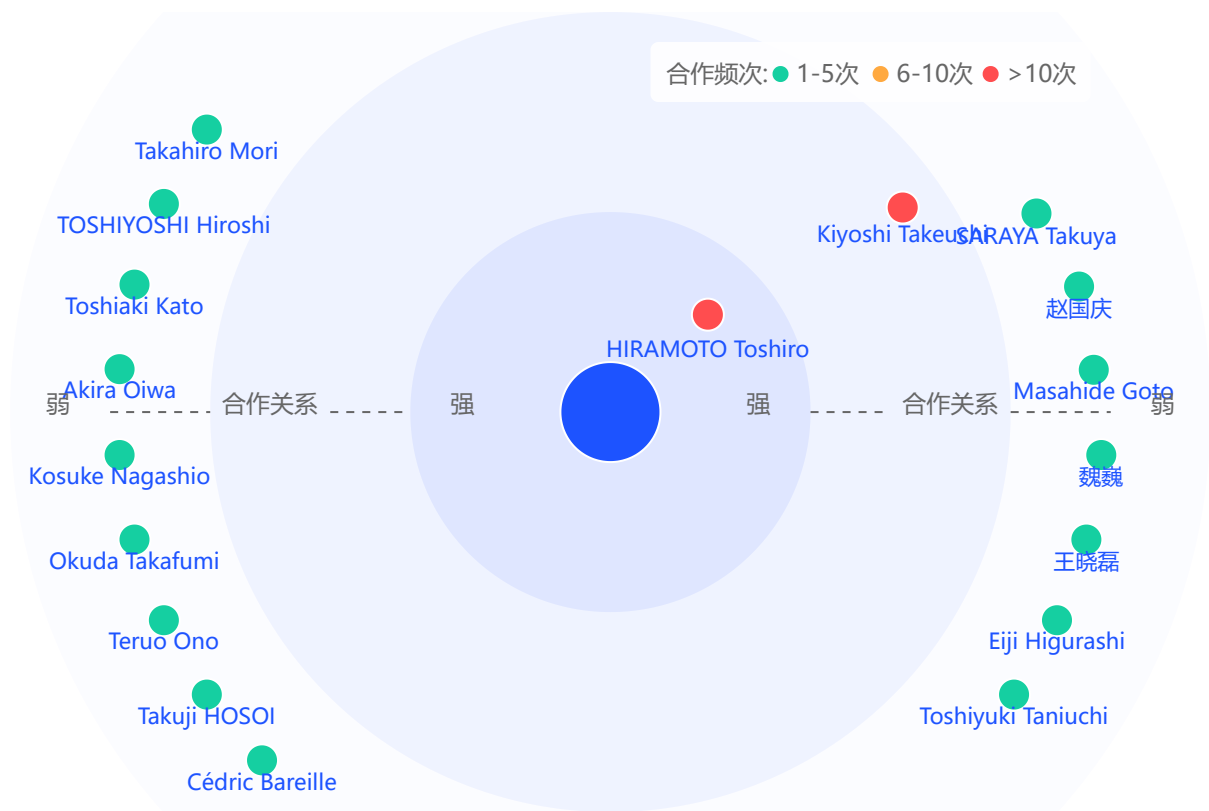
JOURNAL OF APPLIED PHYSICS

Kobayashi, Masaharu; Hiramoto, Toshiro

被引用次数: 67

2008

学术合作



HIRAMOTO Toshiro

东京大学

合作次数: 48

Kiyoshi Takeuchi

东京大学

合作次数: 19

SARAYA Takuya

东京大学

合作次数: 4

赵国庆

山东大学

合作次数: 2

Masahide Goto

东京大学

合作次数: 2

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